

Katla Sathwik

AI/ML Engineer | Deep Learning & Audio ML | Embedded Systems |
Voice AI | Python, C/C++

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Professional Experience

03/2025 – 06/2025

AI/ML Intern

Brainwave Matrix (Virtual) ✕

Fake News Detection System: Built an AI-powered system using Python and NLP to detect fake news with 90%+ accuracy.

- Developed and optimized **machine learning models** (Logistic Regression, LSTM) for reliable classification.
- Implemented **feature extraction** techniques like TF-IDF and **sentiment analysis**.
- Created a **Flask-based web app** for real-time news authenticity checks.
- **Empowered users** to identify misinformation and **combat fake news** effectively.

Tech Stack: Python, NLP, scikit-learn, TensorFlow, Flask

Skills

AI/ML & Deep Learning:

- **Machine Learning:** Scikit-learn, XGBoost, Feature Engineering, EDA, Hyperparameter Tuning, Model Evaluation
- **Deep Learning:** PyTorch, TensorFlow, Keras, CNNs, LSTMs, Transformers (BERT, GPT), U-Net, Seq2Seq, Attention Mechanisms, Encoder-Decoder
- **NLP & LLMs:** HuggingFace Transformers, LangChain, NLTK, SpaCy, Prompt Engineering, Tokenization, Fine-tuning, Semantic Search, Resume Parsing, Semantic Kernel

Programming Languages:

C, C++, Python, Java

Data Science & Analytics:

Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms:

VS Code, Jupyter Notebook, Postman, Linux, Git

Web Development & Cloud:

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Flask, Django, FastAPI, Node.js
- **Databases:** MySQL, MongoDB, Firebase
- **DevOps/Cloud:** Docker, Git, GitHub, API Development & Deployment

Computer Vision:

OpenCV, YOLO, MediaPipe, Pose Estimation, Real-time Image Processing, Data Augmentation

Generative AI:

Stable Diffusion, LLMs, Prompt Engineering

Projects

PlantDoc – AI Disease Detection for Farmers ✕

Major Project | 12/2024 - 01/2025

- Developed an AI-powered system to detect plant diseases from leaf images, reducing crop loss by 25% through early detection.
- Implemented **deep learning models (CNNs)** for high-accuracy image classification (95%+), integrated with mobile and web platforms.

- Conducted extensive **dataset augmentation and preprocessing** to enhance model robustness across real-world conditions.
- Deployed the trained model via a **lightweight mobile-friendly app**, enabling real-time disease diagnosis even in low-connectivity areas.
- Initiated experiments in **model pruning and quantization** to reduce model size and inference latency for potential **embedded deployment**.
- Integrated **OpenCV** pipelines for real-time image capture and preprocessing, enhancing field usability.

Tech Stack: Python, TensorFlow/Keras, OpenCV, Flask, Flutter, Docker, ONNX (planned)

CareerForge – AI-Powered Career Platform

Personal Project | 05/2025 - present

- Designed and developed an AI-driven platform that personalizes career roadmaps and automates job applications, enhancing user job search efficiency by 30%.
- Built a modular **resume parser and job matcher** using transformer-based models in **PyTorch** and **HuggingFace**, optimized for scalability.
- Implemented ML algorithms for dynamic skill assessment and real-time job matching, increasing relevant job suggestions by 25%.
- Engineered interactive modules for **coding practice** and **mock interviews**, improving user interview readiness.
- **Containerized LLM pipelines** using **Docker** and deployed via Flask with MongoDB backend for high availability.
- Leveraged advanced **NLP techniques** for parsing and customizing resumes at scale, streamlining bulk job applications.
- Initiated exploration of **quantization and ONNX export** for model optimization, preparing for edge/embedded inference deployment.

Tech Stack: Python, PyTorch, TensorFlow, HuggingFace, React, Node.js, Flask, MongoDB, Docker, ONNX (learning)

BargainBot – E-commerce Negotiation AI

Personal Project | 03/2025 - 04/2025

- Engineered an AI-powered chatbot automating price negotiations on e-commerce platforms, increasing user savings by 20%.
- Fine-tuned GPT-style models and deployed **agentic workflows** using **LangChain** to handle dynamic conversation flows.
- Built advanced **NLP modules** using Transformers and **SpaCy** to accurately interpret customer intent and generate persuasive counteroffers.
- Designed real-time analytics layer to adjust negotiation strategies, boosting conversion rates.
- Developed a **negotiation strategy engine** leveraging Transformers and attention-based architectures for responsive interaction modeling.
- Optimized backend deployment with **Docker containers**, exploring **model quantization** techniques for future hardware-accelerated inference.
- Deployed as both a web app and mobile app for cross-platform accessibility.

Tech Stack: Python, NLP (SpaCy, Transformers), TensorFlow, React, Node.js, Docker, LangChain

Sign Language Action Detection

Personal Project | 10/2024 - 11/2024

- Developed a real-time computer vision system for detecting and classifying sign language gestures, improving accessibility for the hearing-impaired.
- Designed and trained **CNN + LSTM** based deep learning pipeline, achieving 92%+ classification accuracy on video gesture datasets.
- Integrated **pose estimation, skeletal tracking, and image augmentation** techniques for model robustness under diverse conditions.
- Deployed the trained model on a **mobile application** using a Flask backend, enabling offline usage.
- Began exploration of **inference optimization** through **TensorFlow Lite conversion**, reducing model size for deployment on resource-constrained devices.
- Gained exposure to **real-time video processing pipelines** using OpenCV and MediaPipe.
- Proposed extension of project to **voice command recognition**, bridging to multimodal AI applications.

Tech Stack: Python, OpenCV, TensorFlow/Keras, MediaPipe, Flask, Flutter, TensorFlow Lite (in progress)

Bootcamp

Deep Learning Bootcamp – Attendee

Massachusetts Institute of Technology (MIT) | 01/2025

- Attended intensive lectures and software labs covering **CNNs, RNNs, GANs, LLMs, and Reinforcement Learning**.
- Worked on hands-on projects like **Music Generation, Facial Detection, and LLM Fine-tuning**.
- Participated in final project presentation, applying **advanced deep learning concepts to real-world problems**

Education Credentials

Vaagdevi Engineering College

BTech : Electronics and Communication Engineering

Dec 2021 - Jun 2025 | Warangal, India

CGPA : 7.5/10

Awards & Achievements

- 🏆 **Runner-up**, *State-Level Technical Seminar* on “**Emerging Trends in Artificial Intelligence**”, 2022
- 🏆 **Silver Medalist**, "General Knowledge Olympiad", 2019

Interests

- Startups & Entrepreneurship
- Cricket & Fitness
- Tech Reading